Competitive TCG Data Mining Project

M3: Complete Implementation

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Implementation Status

The complete implementation of the TCG data mining project begins with automated data ingestion from the “Limitless TCG API**”.** The data then undergoes preprocessing, where certain components are removed to isolate the core components of each decklist. Eventually, the data ends up as a matrix filled with card choices and frequencies. The data pipeline is fully operational and integrated within a Jupyter Notebook environment. This implementation analyzes the data using four distinct data mining techniques. First, “FP-Growth” was employed for association rule mining to determine which “Immutable Cores” (cards that are commonly played together) existed by observing which cards were commonly played together. Second, **“**Principal Component Analysis” (PCA) was utilized to compress the card matrix into a two-dimensional “Metagame Map” to visualize archetype clusters. To differentiate between standard and rogue strategies, the “Local Outlier Factor” (LOF) provided density-based anomaly detection, flagging these rogue strategies. Finally, a “Decision Tree Classifier” served as an interpreter, translating the coordinates from the last two techniques into rules that could be understood. Each of these components is currently functional and producing stable, reproducible outputs that directly address the project's goals.

Analytical Choices

A primary analytical challenge in this milestone was addressing Data Noise and collinearity within the decklist structures. Initial iterations of the FP-Growth algorithm were frequently dominated by forced evolution lines, such as the requirement that evolution cards be linked to their parent cards. For example, Dragapult ex must evolve from Drakloak, so in a Dragapult ex deck, you will always run the two together, no matter what. This would clutter the table with obvious conclusions, which obscured more nuanced synergies. To resolve this, I set a confidence threshold of 0.95 and added a "Disjoint Itemset" loop. This approach effectively filtered out predictable evolution lines and ensured the discovery of six distinct, non-overlapping engines rather than redundant representations of universal staples. Furthermore, for anomaly detection, LOF was prioritized over simple distance-based metrics. Because the Pokémon TCG metagame consists of multiple high-density clusters, LOF’s density-based approach was necessary to accurately identify rogue archetypes sitting in isolated regions of the map, even when they share close proximity to established meta clusters.

Results

The results from the data mining pipeline reveal a highly polarized metagame governed by a specific hierarchy of core cards. The Decision Tree classification identified “Buddy-Buddy Poffin” as the primary axis of the 2026 format, mathematically splitting the competitive field into two major strategic families. The split is because “Buddy-Buddy Poffin” is a card that benefits slower, evolution strategies. This card's existence makes those decks split off from faster strategies utilizing non-evolution Pokémon. Beyond mere classification, the correlation between archetype popularity and tournament performance emerged from a comparative analysis of placement data. While standard meta decks achieved an average tournament rank of 26.5, rogue outliers suffered a significant performance penalty, averaging a rank of 42.1. This 15.6-spot gap provides empirical evidence that while rogue strategies offer high strategic variance and a surprise factor, the established meta cores provide a mathematically superior performance floor for consistent competitive success.

Discovery Questions Progress

Question 1: Can unsupervised clustering algorithms accurately reconstruct competitive deck archetypes from raw card vectors, and do they reveal distinct sub-variants missed by manual classification?

The implementation demonstrates that dimensionality reduction through Principal Component Analysis (PCA) effectively reconstructs the metagame geography, condensing over 600 card variables into a 2D map of distinct archetype "islands." The accuracy of this reconstruction is validated by the Decision Tree Classifier, which translated these abstract clusters back into recognizable strategic rules, such as the split between "Poffin-reliant" and "Poffin-independent" archetypes. Furthermore, the map revealed distinct sub-variants—smaller clusters drifting near major archetypes —that represent specific tech choices a manual, top-down classification system might otherwise overlook as mere noise.

Question 2: What frequent itemsets characterize the top-performing decks, and can we mathematically define the boundary between a deck’s “Immutable Core” and its “Flex Slots”?

By utilizing FP-Growth with a refined confidence threshold of 0.85, the project successfully defined the "Immutable Core" as high-lift itemsets that persist across the majority of successful tournament entries. Mathematically, the boundary is established by the Lift and Confidence scores; cards that appear in these frequent itemsets represent the structural foundation of the deck, while those falling below the support threshold are identified as "Flex Slots." This distinction allows for a clear separation between mandatory engine components and the high-variance card choices players use to adapt to local metagame shifts.

Question 3: Which high-ranking decks are statistical outliers relative to the established meta clusters?

The integration of LOF and tournament placement data specifically identified the "Rogue" archetypes that defy standard meta clusters. While the majority of top-performing decks reside in the high-density PCA islands, the analysis flagged a subset of high-ranking outliers, represented by red data points on the map, that achieved competitive placements despite using statistically unique card combinations. However, the final performance comparison revealed that while these outliers exist, they suffer a 15.6-spot average placement penalty compared to the standard meta decks, proving that while outlier success is possible, it is significantly less consistent than adhering to the established "Immutable Core."

M4 Plan

The final phase of the project, Milestone 4, will focus on refining the presentation and depth of these existing findings. One primary objective is the transition from raw Matplotlib outputs to professional-grade visualizations with consistent color-coding and enhanced legends to improve accessibility. Additionally, I plan to explore whether the “Match Win Percentage" metric offers a different perspective from "Tournament Placement," adding another layer to the performance analysis. The ultimate goal for M4 is to synthesize these technical outputs into a cohesive "State of the Meta" report that effectively communicates the game's underlying structure.